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Should We Lose Sleep Over Sleep Disturbances After Sports-Related Concussion? A Scoping Review of the Literature

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Objective: A single, severe traumatic brain injury can result in chronic sleep disturbances that can persist several years after the incident. In contrast, it is unclear whether there are sleep disturbances after a sports-related concussion (SRC). Considering growing evidence of links between sleep disturbance and neurodegeneration, this review examined the potential links between diagnosed SRCs and sleep disturbances to provide guidance for future studies. **Methods:** The scoping review undertook a systematic search of key online databases (Scopus, MEDLINE, SportDiscus, and Web of Science) using predetermined search terms for any articles that examined sleep after concussion. A screening criterion using agreed inclusion and exclusion criteria was utilized to ensure inclusion of relevant articles. **Design:** This scoping review is guided by the PRSIMA Scoping Review report. **Results:** Ten studies met the inclusion criteria, reporting on 896 adults who had experienced an SRC. Comparison with 1327 non-SRC adults occurred in 8 studies. Nine studies subjectively examined sleep, of which all but one study reported sleep disturbances after an SRC. Three studies objectively measured sleep, with 2 studies indicating large coefficients of variation of sleep duration, suggesting a range of sleep responses after an SRC. The only study to examine overnight polysomnography showed no differences in sleep metrics between those with and without an SRC. No studies examined interventions to improve sleep outcomes in people with concussion. **Conclusions:** This scoping review indicates preliminary evidence of sleep disturbances following an SRC. The heterogeneity of methodology used in the included studies makes consensus on the results difficult. Given the mediating role of sleep in neurodegenerative disorders, further research is needed to identify physiological correlates and pathological mechanisms of sleep disturbances in SRC-related neurodegeneration and whether interventions for sleep problems improve recovery from concussion and reduce the risk of SRC-related neurodegeneration. **Key words:** *athletic injuries*, *brain concussion*, *craniocerebral trauma*, *chronic traumatic encephalopathy*, *sleep*, *sleep-wake disorders*

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SPORTS-RELATED CONCUSSION (SRC) is a form of mild and/or minimal traumatic brain injury (TBI)[1] that is an ever-present concern for participants in many sports.[2] Participants in collision sports, such as ice hockey,[3] American football,[4] and both rugby union and rugby league,[5,6] are particularly at risk of experiencing an SRC. Furthermore, many athletes report experiencing multiple concussions throughout their career.[7,8] Yet, the incidence and impact of SRCs may be underestimated, as 40% to 60% of concussion events go undiagnosed or unrecognized.[9]

A concussion results from biomechanical trauma to the head, such as a direct impact, or a sudden indirect acceleration, deceleration, and/or rotation of the head with or without the loss of consciousness.[10] The biomechanical trauma causes shearing and/or tearing neuronal axons, leading to a neurometabolic cascade that results in altered resting states and brain activation patterns.[11,12]

The chronic adverse health impacts of repeated SRCs are rapidly becoming a public health concern.[13] Acutely, concussion can result in a broad spectrum of physical, emotional, and cognitive symptoms.[14] Furthermore, there is growing evidence that those exposed to repeated concussions may experience "chronic traumatic encephalopathy" (CTE) in later life.[15] CTE is typified by the excess accumulation of amyloid-β and τ-proteins, which are implicated in neurodegenerative disorders including Alzheimer disease, Parkinson disease, and amyotrophic lateral sclerosis.[16,17] Yet, how repeated SRCs and the development of CTE are associated still needs to be established.[18] One possible explanation implicates sleep disturbances, as even a single night of sleep deprivation results in the accumulation of amyloid-β and τ-protein.[19,20] Current evidence indicates that disruptions of slow-wave sleep (SWS) interrupt the clearance of amyloid-β and τ-protein by the glymphatic system, resulting in accumulation of amyloid-β and τ-proteins.[21] Unsurprisingly, sleep disturbances such as irregular sleeping schedule and irregular sleeping hours, as well as clinical sleep disorders such as obstructive sleep apnea (OSA), and insomnia, are considered primary risk factors for neurodegenerative disorders due to the disruption of SWS.[22–25]

Between 30% and 66% of survivors of TBI report sleep disturbances, such as trouble initiating and/or maintaining sleep, or symptoms of hypersomnolence, such as sleeping for excessive periods of time (ie, sleeping more than the recommended 7-9 hours a night), which persist for several years after the initiating event.[26] While sleep disturbances are widely recognized as symptoms of concussion,[27,28] our understanding of the relationship between sleep and concussion is much more limited. Thus, a scoping review on the association of SRCs with sleep in adults was undertaken to establish the extent of research in the area, with the aim of establishing a preliminary evidence-based synthesis of the topic, and to provide directions for future research in the area.

METHODS

This article follows the PRISMA Extension for Scoping Reviews (PRISMA-ScR) guidelines.[29] To our knowledge, there has been no literature or systemic reviews examining the effect of SRCs on sleep; meaning, a scoping review was conducted. A scoping review would identify the current scope of research, and therefore any knowledge gaps, and establish the methodologies used all while still undergoing a rigorous and transparent methodology.[30–32] Scoping reviews are not eligible for PROSPERO registration.

Our primary research question was as follows: "Are there sleep disturbances after experiencing a sports-related concussion?" We limited the scoping review to adults (18 years and older) due to the differences in sleep patterns and sleep disorders between adolescents and adults.[33,34] Furthermore, to assess the characteristics of the methodologies used, the secondary research question was as follows: "What are the methodological characteristics of studies examining the effect of concussion on sleep, namely, how was sleep assessed, how was concussion assessed, the study design, and populations used?"

Scopus, MEDLINE (via PubMed), SportDiscus, and Web of Science databases were systematically searched from inception to identify potentially relevant studies. Utilizing key words around SRC, sports, and sleep, as well as using Boolean operator terms, the search terms were (("concus*" OR "mild traumatic brain" OR "mTBI") AND ("sleep" OR "circadian" OR "insomnia" OR "sleep disord*" OR "sleep dist*" OR "obstructive sleep") AND ("athletes" OR "rugby" OR "football" OR "AFL" OR "NFL" OR "contact sport" OR "collision sport")). Where available, filters to remove review articles, editorials, and conference abstracts, as well as non–English language documents, were applied. An initial search was undertaken in November 2019, with a second search completed in September 2020.

English language, prospective and retrospective cohort studies, and randomized controlled trials (RCTs) that examined SRC-induced changes to sleep or interventions to improve sleep after an SRC were eligible for inclusion. Review articles, editorials, and conference abstracts were excluded.

Eligible studies included adults (older than 18 years) who suffered a concussion during athletic performance. Concussion was required to be identified by professional diagnosis, a concussion assessment tool (eg, Sports Concussion Assessment Tool [SCAT]), or

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explicit participant declaration of a previous medically diagnosed concussion. Studies examining concussion in children (younger than 18 years), animal models of concussion, or concussion acquired outside athletic performance were excluded.

Sleep outcomes included any subjective or objective measures of sleep. Eligible subjective measures included standardized questionnaires, such as the Pittsburgh Sleep Quality Index (PSQI),[35] or individual sleep-related items from a validated diagnostic tool, such as the SCAT,[36] or specific questions pertaining to sleep but not part of any validated questionnaire. Eligible objective sleep assessments were either actigraphy or polysomnography (PSG). Actigraphy is a device worn on the body that uses accelerometery to determine sleep by extended periods of nonmovement.[37] PSG collects neurophysiological data to determine total sleep time, wake after sleep onset, sleep efficiency, and how much each stage of sleep (stages 1, 2, SWS, and rapid eye movement sleep) contributes to overall sleep for a specific night.

Eligibility of the studies were evaluated independently by 2 authors (D.J.S. and A.A.), with disagreements between reviewers arbitrated by a senior author (L.H.). After removing duplicates, the titles and abstracts of all studies were screened. Those considered potentially eligible were read in full to determine inclusion. In studies with insufficient published information, the corresponding author of that article was contacted.

For articles meeting all inclusion and no exclusion criteria, 2 of the authors (D.J.S. and A.A.) extracted data based on the following categories:

1. Primary author and year of publication;
2. Number and gender of concussed and control participants;
3. Mean age of the entire cohort, and all concussed and all controls;
4. Study design
   i. Within subjects (pre/posttest assessment), between participants (separate control group), or observational (no control group);
   ii. Prospective (participant recruited either to monitor incidence of concussion or were recently concussed, and were still recovering), and/or
   iii. Retrospective (participants who previously experienced a concussion, and were no longer symptomatic), and/or
   iv. RCT (induced sleep change or treated sleep complaint);
5. Concussion specifics,
   i. diagnostic method and criteria (physician, assessment tool, self-report);
   ii. number of concussions;
   iii. time between concussion and sleep measure;
6. Sleep measurement: Subjective and objective; and
7. Outcome/information: Descriptive (eg, means and standard deviations) and inferential statistics (eg, *P* values).

Extracted data were synthesized in a standardized table (see Table 1). As per scoping review guidelines, risk of bias and quality assessments were not undertaken.[29]

RESULTS

In total, 10 of 292 potentially relevant articles met study inclusion criteria, which resulted in 896 athletes who had experienced an SRC (involved in women's competition = 259).[38–47] Reasons for exclusion are outlined in the PRISMA diagram (see Figure 1). Investigators of 2 studies were contacted for further information. Hoffman et al[43] explained how sleep duration was calculated for their study, and Davis-Hayes et al[40] clarified the sleep-specific question of the Concussion Resolution Index used in their study.

Assessment of concussion

Physician diagnosis of SRC occurred in 6 studies.[39,40,43–46] Two studies utilized participant report of a physician diagnosis,[41,42] while the remaining 2 studies utilized self-report of SRC.[38,47]

The clinical definitions of concussion are outlined in the Table 1. Five studies[39,40,42,43,46] utilized the definition created by the Concussion in Sport Group (which created, and update, the SCAT questionnaire).[36] Two studies used definitions that were bought about by expert reviews of the literature. Participants from the Hoffman et al[43] study were recruited as part of the Concussion Assessment, Research and Education (CARE) Consortium.[48] The CARE Consortium utilized the definition created by a Panel of Technical Experts.[49] Gosselin et al[41] utilized the definition from the World Health Organization "Collaborating Centre for Traumatic Brain Injury."[50] Two studies appeared to use "self" created definitions of concussion,[45,47] while the remaining study did not report the definition of concussion used,[38] only symptomatology from the Rivermead Postconcussion Questionnaire.[51]

Assessment of sleep and/or sleep-related daytime dysfunction

Three studies utilized validated sleep questionnaires, with Gosselin et al[41] and Hoffman et al[44] administering the PSQI and the Epworth Sleepiness Scale (ESS), while Hutchison et al[45] utilized the Stanford Sleepiness Scale. Three studies utilized individual sleep items from concussion screening tools, being the SCAT[39,44] and the Headminder Concussion Resolution Index.[40] Gouttebarge et al[42] examined sleep items from the Patient Reported Outcomes Measurement Information System (PROMIS), a patient health monitoring system.

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TABLE 1 *Synthesis of the extracted data from the included studies*

Study	Participants	Mean age, y	Study design	Concussion specifics	Sleep measurement	Outcome
Blake et al (2019)[38]	Cohort = 82 (*f* = 52) Concussion = 42 (*f* = N/A) *Sports* Baseball Soccer Softball Ultimate frisbee Athletics Rugby Lacrosse Volleyball Basketball Field hockey Other	Cohort = 20.3 ± 1.5 Concussion = NA	Retrospective Between participants	Dx: Self-report 40/82 report 0 concussions 29/82 report 1-2 concussions 13/82 reported 3+ concussions Concussion definition: NA Symptoms: Rivermead Postconcussion Questionnaire	Subjective: Self-report of common symptoms (not validated); difficulty falling asleep, restlessness, sleeping too much, sleeping too little, difficulty staying asleep, fatigue Sleep measures relative to most recent concussion: NA	Differences between groups ($P = .028$). Post hoc analysis showed 3+ prior concussions significantly more sleep disturbances vs 0 concussions ($P = .012$). No other differences reported. Prevalence of sleep disorders for each concussion group was not reported.
Churchill et al (2017)[39]	Control 35 (*f* = 19) Concussed 35 (*f* = 19) *Sports* Volleyball Hockey Soccer Football Rugby Basketball Lacrosse	Concussed = 20.3 ± 2.2 Control = 20.3 ± 1.7	Prospective Between and within participants	Dx: Physician Definition: Concussion in Sport Group	Subjective: Fatigue and sleep symptom questions of the SCAT3; fatigue/low energy, drowsiness, trouble falling asleep Sleep measures relative to most recent concussion: 3.6 ± 1.2 d	No difference in fatigue and sleep symptom severity from pre- to postinjury for the concussed participants ($P = .36$), nor differences between concussed postinjury and to nonconcussed controls ($P = .16$).
Davis-Hayes et al (2017)[40]	Cohort = 1200 (*f* = 378) Concussed = 228 (*f* = 88) *Sports* Football Wrestling Soccer Basketball Field hockey Lacrosse Baseball Volleyball Softball	Cohort = 19.0 ± 1.0 Concussed = N/A	Prospective (retrospective analysis of cohort) Between participants	Dx: Physician Definition: Concussion in Sport Group	Subjective: Sleep disturbance questions from the "Headminder Concussion Resolution Index System"[a] Sleep measures relative to most recent concussion: <24 h	29% males and 42% of females reported sleep disturbances. Actual sleep disturbances not reported. Females incidence was significantly higher than for males ($P = .048$). *Symptomatology only examined after the first concussion, meaning no comparison with nonconcussed controls.*

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TABLE 1 *Synthesis of the extracted data from the included studies* (Continued)

Study	Participants	Mean age, y	Study design	Concussion specifics	Sleep measurement	Outcome
Gosselin et al (2009)[41]	Control = 11 (*f* = 4) Concussed = 10 (*f* = 3) *Sports* Soccer Football Hockey Rugby Skate (*described in the text as part of a "skate team"*)	Control = 22.6 ± 2.4 Concussed = 24.3 ± 6.1	Retrospective Between participants	Dx: Self-report of physician diagnosis Definition: World Health Organization Collaborating Centre for Traumatic Brain Injury	Objective: PSG, which measured total sleep time (min), sleep-onset latency (min), wake after sleep onset (min), arousal index, sleep efficiency (%), individual sleep stage (N1, N2, deep, and REM)%, REM onset (min), REM efficiency. Subjective: Sleep quality (PSQI), daytime sleepiness (ESS). Sleep measures relative to most recent concussion: 1-11 mo	No group differences in any PSG measures (*P* values not reported). Concussed participants had significantly worse quality sleep (PSQI: 7.7 ± 3.8 vs 3.5 ± 2.3, *P* = .006), no difference in sleepiness (9.9 ± 6.1 vs 9.6 ± 6.1, *P* value not reported).
Gouttebarge et al (2017)[42]	Cohort = 576 Concussion = 363 *All participants were male.* *Sports* Football Ice hockey Rugby	Control = 37 (20-50) Concussed = N/A	Retrospective Between participants	Dx: Self-report of physician 0 concussions 110/576 1 concussion 174/576 2-3 concussions 158/576 4-5 concussions 84/576 6+ concussions 50/576 Definition: Concussion in Sport Group	Subjective: Sleep disturbance questions of the PROMIS *In the past 7 d,* • *My sleep quality was…* • *My sleep was refreshing* • *I had a problem with my sleep* • *I had difficulty falling sleep* Rankings: Very poor, poor, fair, good, very good Sleep measures relative to most recent concussion: NA *All participants had retired 7 ± 5 y ago*	Compared with 0 concussion controls, odds ratios (95% confidence intervals) of reporting sleep disturbances per incidence of concussion (*P* values not reported); 1 concussion = 1.8 (0.8-3.9) 2-3 concussions = 1.9 (1.0-3.6) 4-5 concussions = 2.0 (1.0-4.2) 6+ concussions = 5.2 (2.4-11.2). *Details/scored of questions not provided.*

(*continues*)

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TABLE 1 *Synthesis of the extracted data from the included studies* (Continued)

Study	Participants	Mean age, y	Study design	Concussion specifics	Sleep measurement	Outcome
Hoffman et al (2020)[43]	Cohort (all concussions) = 151 ($f = 66$) *Sports* NA[b]	Cohort = 19.3 ± 1.2	Prospective (retrospective analysis of cohort) Within participants	Dx: Physician Definition: Panel of Technical Experts (Ref)	Subjective: Self-report changes in sleep duration (postconcussion total sleep time – baseline total sleep time)[c] Sleep measures relative to most recent concussion: within 2 d	81/151 reported longer sleep (2.2 ± 1.5 h) 35/151 reported no change in sleep (0.1 ± 0.3) 35/151 reported shorter sleep (2.4 ± 1.2). "Short sleepers" indicated significantly more concussion symptoms within first 4 d after concussion compared with both no difference and long sleepers (*P* values not reported). No significant differences with the first 4 d existed between "normal sleepers" and "long sleepers." Time to asymptomatic, and return to play, was not significantly different between any groups. *NB: 3+ concussions excluded from analysis.*
Hoffman et al (2019)[44]	Control = 20 ($f = 12$) Concussed = 20 ($f = 12$) *Sports* NA	Control = 20.4 ± 1.4 Concussed = 20.0 ± 1.5	Prospective Between participants	Dx: Physician Definition: Concussion in Sport Group	Objective: Actigraphy Subjective: Sleep diary (used as adjunct for actigraphy only), sleep quality (PSQI), daytime sleepiness (ESS), sleep symptom severity ([SCAT3]: drowsiness + trouble falling asleep). Sleep measures relative to most recent concussion: <3 d	Compared with nonconcussed participants Concussed participants experienced significantly longer sleep-onset latency ($P = .017$) compared with healthy controls. Post hoc indicated sleep-onset latency at 2-3 d postinjury significantly longer in the concussed group ($P = .002$).

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TABLE 1 *Synthesis of the extracted data from the included studies* (Continued)

Study	Participants	Mean age, y	Study design	Concussion specifics	Sleep measurement	Outcome
						Concussed participants demonstrated significantly increased variability of total sleep time ($P = .044$) and wake after sleep onset ($P = .017$) compared with healthy controls. Concussed participants recorded significantly worse sleep quality (PSQI: 7.8 ± 3.0 vs 4.2 ± 1.8, $P < .001$) and daytime sleepiness (12.6 ± 5.3 vs 8.5 ± 4.4, $P < .001$). SCAT3 sleep symptom significantly higher in concussed participants (4.4 ± 3.2 vs 0.8 ± 1.4, $P = .014$) compared with healthy control.
Hutchison et al (2017)[45]	Control = 26 (*f* = 10) Concussed = 26 (*f* = 10) *Sports* Football Soccer Rugby Hockey Basketball Volleyball Lacrosse Baseball	Control = N/A (no significant difference to concussed) Concussed = 21.0 ± 2.5	Prospective Between participants	Dx: physician Definition: "*(1) Observed or reported acceleration/ deceleration of the head,* *(2) Any observed alteration in mental status, and/or* *(3) Signs such as confusion, vacant stare, poor coordination, difficulty concentrating, poor balance, and/or* *(4) Any self-reported symptoms such as headache, loss of consciousness, nausea, balance problems, or difficulty reading or concentrating.*"	Subjective: Sleepiness (SSS) Sleep measures relative to most recent concussion: (*a*) Within 7 d (symptomatic) (*b*) Asymptomatic (days not reported) (*c*) 7 d post–return to play	Concussed significantly worse sleep symptoms (*P* values reported only as <.05) during symptomatic phase compared with controls (*SSS scores not reported).* No differences in sleep during asymptomatic and post–return to play.

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TABLE 1 *Synthesis of the extracted data from the included studies* (Continued)

Study	Participants	Mean age, y	Study design	Concussion specifics	Sleep measurement	Outcome
Raikes and Schaefer (2016)[46]	Control = 10 ($f = 5$) Concussed = 7 ($f = 4$) *Sports* NA	Control = 24.1 ± 1.9 Concussed = 22.7 ± 1.2	Prospective Between participants	Dx: Physician Definition: Concussion in Sport Group	Objective: Actigraphy Sleep measures relative to most recent concussion: <4 d	No significant differences between concussed and nonconcussed participants for total sleep time, sleep efficiency, and wake after sleep time ($P > .05$ for all variables).
Vagnozzi et al (2008)[47]	Cohort (all concussions) = 14 ($f = 5$) *Sports* NA	Concussed = 26.5 (20-35) (*NB individual ages reported, self-calculated)*	Prospective Observation	Dx: Self-report Definition: "*A traumatically induced alteration in mental status, not necessarily with loss of consciousness*"	Subjective: Self-report sleep symptoms Sleep measures relative to most recent concussion: <3 d	4/14 reported sleep problem 2 reported trouble falling asleep 2 reported sleeping too long

Abbreviations: Dx, diagnosis; ESS, Epworth Sleepiness Scale; NA, not available; PROMIS, Patient Reported Outcomes Measurement Information System; PSG, polysomnography; PSQI, Pittsburgh Sleep Quality Instrument; ref, reference; REM, rapid eye movement; SCAT3, Sports Concussion Assessment Tool (version 3); SSS, Stanford Sleepiness Scale.

[a]The authors have been contacted to clarify the questions asked in this concussion assessment tool.

[b]Data were from the Concussion, Assessment, Research and Education (CARE) Consortium, which has recruited more than 41 000 male and female National Collegiate Athletic Association athletes and service academy cadets.

[c]In their article, they describe *"Using the SCAT3, self-reported number of hours of sleep from the previous night was obtained as baseline and within 24-48 hours PI (post-injury)"*; however the SCAT3, nor any versions of the SCAT, measures self-reported sleep duration. Personal correspondence between primary authors clarified that sleep duration was measured in addition to questions asked during the symptom checklist of the SCAT.

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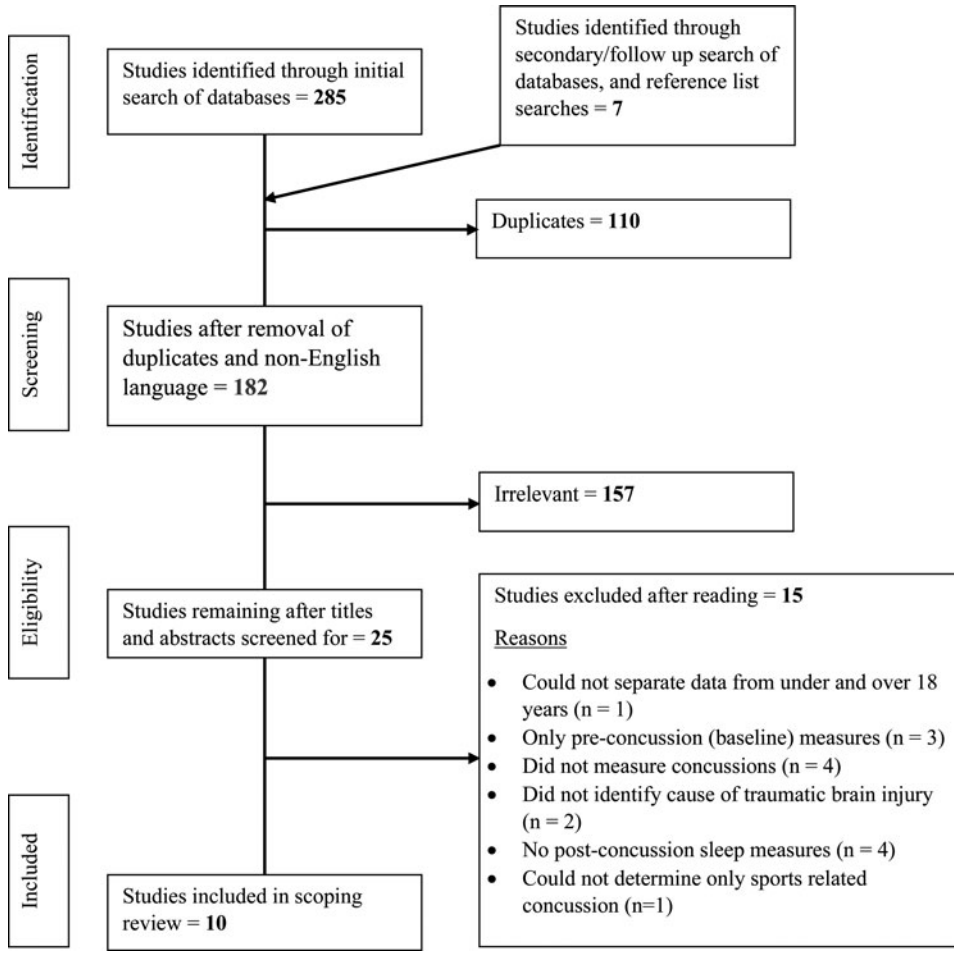

Figure 1. PRIMSA flow diagram of literature search and selection.

Three studies utilized nonvalidated and unstandardized questionnaires. Hoffman et al[43] examined perceived changes in sleep duration in the 48 hours postinjury compared with before injury, as in more, less, or the same sleep duration. Blake et al[38] and Vagnozzi et al[47] collected self-reported sleep disturbances, such as trouble initiating sleep or sleeping too long.

Three studies utilized objective sleep measures. Two utilized actigraphy,[44,46] while the other utilized PSG.[41] Two of these studies also examined subjective measures using the ESS and the PSQI.[41,44]

Time between concussion and sleep assessment

Seven of the studies assessed sleep within a week of the study's participants sustaining an SRC,[39,40,43–47] with 2 of these studies undertaking longitudinal assessments of sleep.[43,45] One study examined sleep between 1 and 11 months after the participant's concussion,[41] while another examined sleep symptoms several years after retiring from sports, although did not mention when their participant's last concussion occurred.[42] Another study also did not report the time between the participant's sustaining an SRC and the sleep assessment.[38]

Associations of concussion and sleep

Overall, 8 of the 9 studies examining subjective sleep measures described some form of sleep disturbance occurring after SRC.[38,40–47] Two studies examined changes in sleep duration.[43,47] Hoffman et al[43] reported that half of participants in their cohort who experienced SRC were sleeping for 2 hours longer than what they did prior to the concussion, whereas just under a quarter reported sleeping 2 hours less than what they did prior to concussion. The remaining participants reported no changes in sleep duration.[43] Vagnozzi et al[47] reported that 4 of 14 participants reported sleep disturbances, with 2 participants reporting trouble falling asleep while the other 2 reported sleeping more than usual. Two studies demonstrated those who experienced an SRC reported worse sleep quality, assessed by the PSQI, compared with nonconcussed controls.[41,44]

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Three studies examined daytime sleepiness. Two of these studies, one utilizing the ESS and the other utilizing the Stanford Sleepiness Scale, demonstrated those who experienced an SRC reported elevated levels of daytime sleepiness compared with nonconcussed healthy controls.[44,45] In contrast, the remaining study, utilizing the ESS, found no differences in sleepiness between those with SRC and nonconcussed controls[41]; however, data collection for this study occurred at least 1 month after the last SRC, and they did report poorer subjective sleep quality in those who experienced an SRC.

Of the 3 studies that utilized sleep questions from a concussion assessment tool, 2 studies reported sleep disturbances[40,44] while the other did not.[39] Churchill et al,[39] who showed no sleep disturbances after an SRC, combined from the SCAT questionnaire drowsiness, difficulty falling asleep, and "fatigue."[39] Fatigue, however, is not considered a sleep disturbance.[52] In contrast, Hoffman et al,[44] who also used the SCAT questionnaire, only combined the drowsiness and difficulty falling asleep.

Studies using objective sleep data also reported conflicting results. Using actigraphy, Hoffman et al[44] found those with SRC experienced significantly longer sleep-onset latency as well as greater variability in total sleep-wake time after sleep onset than healthy controls.[44] In contrast, Raikes and Schaefer[46] showed no differences in actigraphy-derived sleep measures in 7 participants with SRC compared with 10 healthy controls. Likewise, Gosselin et al[41] showed no differences in PSG-derived sleep measures, including total sleep time, sleep efficiency, and composition of sleep stages, between 11 participants with SRC and 10 healthy controls.

Three studies examined relationships between concussion metrics and sleep disturbances. According to Hoffman et al,[43] those who reported reduced sleep after a concussion had significantly more concussion symptoms than those who reported more sleep or no changes to sleep. Blake et al[38] reported a significant positive correlation between concussion symptomatology and sleep disturbances. They also found that athletes who experienced 3 or more SRCs reported a significantly higher number of sleep disturbances than those with fewer than 3 SRCs and nonconcussed controls. Gouttebarge et al[42] showed that athletes with a history of 1 to 5 SRCs over their career were approximately twice as likely and those with 6 or more SRCs were 5 times more likely to report sleep disturbances. Furthermore, participants included in this study reported these sleep disturbances several years after sustaining the last concussion. Neither study, however, elaborated on the specifics of the sleep disturbances.

While the majority of studies examined both male and female athletes, only Davis-Hayes et al[40] examined differences in self-reported sleep disturbances between genders after SRC. Female athletes reported significantly higher percentage of sleep disturbances than male athletes. The specific disturbances, however, were not examined.

DISCUSSION

This scoping review identified 10 articles, including 3 large-scale cohort studies, that examined sleep disturbances after SRC in 896 adults. From a mixture of both subjective and objective assessments of sleep, 8 studies concluded that sleep disturbances are common after an SRC,[38,40–45,47] with 7 of these studies assessing sleep within a week of concussion occurring. Such disturbances included poor sleep quality, excessive daytime sleepiness, and perceived changes in sleep duration, with both sleeping longer and sleeping less being reported. Other findings include an apparent dose-response relationship between the number of SRCs experienced and the severity of sleep disturbance symptomatology,[38,42] a potentially long continuation of subjective sleep disturbances after the initial event,[41,42] and the possibility that females are more sensitive to sleep symptoms following an SRC.[40]

Despite a paucity of evidence, changes in sleep duration after an SRC suggest an association between increased sleep duration and improved outcomes. In the studies that examined perceived changes in sleep duration, there was a mix of those reporting increased sleep, decreased sleep, and no change in sleep compared with either nonconcussed controls or prior to concussion.[38,43,47] Furthermore, the 2 studies that objectively measured sleep by actigraphy reported a large coefficient of variations of total sleep time compared with nonconcussed controls. While seemingly contradictory, the difference post-SRC sleep duration allows speculation that sleep is used to help reduce the histological problems of SRC. As previously discussed, concussions lead to the deposition of amyloid-β and τ-protein, which are responsible for propagating CTE.[19,20] The glymphatic system, most active during deep sleep, removes these proteins.[53] Studies in the acute phase of the moderate to severe TBI indicate increased deep sleep,[54,55] suggesting that this is a common outcome. Thus, in people who report sleeping longer than usual after an SRC, it is hypothesized the brain is trying to increase the amount of SWS to maximize the glymphatic drainage. Conversely, for those who report sleeping less than usual after an SRC, they may be susceptible to reduction and/or disruption to deep sleep, which may have the potential to contribute to the development of CTE in later life.

Although the preliminary findings suggest that SRCs result in some form of sleep disturbances, the majority

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of studies utilized subjective measures of sleep, with only 3 studies utilizing validated sleep or sleep-related daytime dysfunction questionnaires.[41,44,45] This is further confounded by the overall heterogeneity of the questionnaires used. The search strategy, as well as the inclusion and exclusion criteria for this scoping review, was intentionally designed to maximize the number of potential articles that could be included; however, we concede this would have contributed to this methodological heterogeneity.

Three studies utilized sleep questions from concussion assessment tools,[39,40,44] which included the only study to show no differences in the prevalence of sleep disturbances between SRC participants and non-SRC controls. Concussion assessment tools have never been validated in determining sleep disturbances. This may explain why Churchill et al,[39] using the SCAT, combined fatigue along with drowsiness and trouble falling asleep to gauge sleep disturbances, whereas Hoffman et al[44] only combined drowsiness and trouble falling asleep. Fatigue is the state of lethargy bought about due to constant mental and/or physical exertion and is distinctly different from sleep disturbances.[56] While sleep interventions alleviate drowsiness, it has no impact on fatigue.[52] Furthermore, the SCAT, which was utilized in 2 studies,[39,44] only asks for "trouble falling asleep." Thus, those sleeping more than normal, an apparent common outcome after an SRC,[43] are not captured, thereby potentially underreporting the prevalence of sleep disturbances.

Three studies examined sleep through "probing" questions, rather than validated questionnaires.[38,43,47] As there were no objective measures of sleep to corroborate what was reported by the participants, the findings of these studies need to be interpreted with caution.

Only one small study examined overnight PSG between those with SRC and nonconcussed controls.[41] Despite SRC participants reporting significantly worse subjective sleep quality assessed by the PSQI, no differences in any of the sleep metrics measured by the PSG, such as sleep duration, objective sleep efficiency, and the composition of sleep stages, were shown. Nevertheless, testing occurred between 1 and 11 months after the last SRC and had relatively few participants (10 participants with SRC, and 11 nonconcussed controls), which makes interpretation of the results difficult. Other studies that compared overnight PSG in moderate to severe TBI in the acute phase with non-TBI populations have shown differences in PSG, namely, increased SWS and increased awakenings from sleep.[54,55] Therefore, it is reasonable to speculate that future studies to examine overnight PSG in the acute phase of an SRC injury will demonstrate differences compared with non-SRC participants.

Clinical implications

No studies in humans have examined interventions to improve sleep in those who have experienced SRC. Only one study has examined the feasibility of implementing such an intervention but has yet to examine whether the intervention improves neural rehabilitation.[57] A study examining TBI in Sprague Dawley rats showed increasing SWS through both behavioral (sleep deprivation resulting in rebound SWS) and pharmacological means (the administration of sodium oxybate), resulted in a reduced concentration of amyloid precursor protein, and improved memory.[58] Therefore, we are cautiously optimistic that interventions designed to improve sleep may result in improved recovery from SRC.

Other factors known to affect sleep, such as mental health problems and medication usage, were poorly controlled in the studies included in this review. Three studies excluded participants who reported mental health problems,[41,44,46] while a further 3 studies collected data on the incidence of mental health problems but no adjustments for these were made in the analyses.[42,43,45] Only one study excluded those who reported using medications known to affect sleep.[44] This is especially pertinent considering more than 10% of athletes may be prescribed sedative/hypnotic medications[59] and more than 5% of athletes have been prescribed opioids.[60] Only 2 studies considered, and subsequently excluded, athletes who reported a previously diagnosed sleep disorder.[44,46] This is despite growing evidence indicating that many athletes suffer from either insomnia-like symptoms, such as difficulty initiating and maintaining sleep, or OSA, which results in nocturnal intermittent hypoxemia.[59,61] Notably, insomnia-like disturbances during the preseason can increase the susceptibility of athletes to SRC during subsequent competition.[62] In addition, both insomnia-like problems and OSA are independently linked with increased levels of amyloid-β and τ-proteins.[63,64] Furthermore, the accumulation of amyloid-β and τ-proteins independently disrupts deep sleep,[65] thereby potentially creating a vicious cycle of sleep disturbance and accumulation of these proteins. Thus, screening athletes for sleep disorders may improve identification of those potentially at risk of sustaining or experiencing poor recovery from an SRC.

Finally, no study has examined the potential role of the apolipoprotein genotype ε4 (APOε4) allele in SRC and sleep. While the APOε4 allele is associated with a reduced number of concussions,[66] those with the APOε4 allele experience a higher risk of τ-proteins and amyloid-β accumulation after head injury.[67,68] Unsurprisingly, the prevalence of the APOε4 allele in those diagnosed with CTE is higher than the population average.[17,69] Importantly, the APOε4 allele increases the prevalence

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of insomnia,[70] OSA,[71] and poor sleep quality.[72] It is possible that athletes who report reduced sleep after an SRC, as well as worse sleep quality long after their last SRC, may have the APOε4 allele. Screening athletes for the APOε4 allele may also facilitate the early identification of those potentially at risk of experiencing poor sleep after an SRC.

Clinical assessment of SRC is usually done with standardized SRC assessment tools such as the SCAT[36] or computerized tools such as Cogstate or ImPact.[73,74] As previously discussed, these tools do not allow for proper assessment of sleep disturbances due to asking superficial questions such as grading of "drowsiness" and "having trouble falling asleep." In the light of a possible association between sleep and recovery from SRC, it is reasonable to recommend adding a question about sleeping more than usual in assessments of SRC. The lack of studies examining the effect of interventions improving sleep after an SRC means definite recommendations for clinical practice cannot be made. Nevertheless, given the beneficial effects of good sleep on health and cognitive function, it is appropriate to recommend that an initial positive finding on sleep disturbances is an indication for a more in-depth sleep assessment with validated subjective sleep assessment tools and/or a sleep diary. Prolonged sleep disturbance may warrant objective assessments with accelerometery and a sleep diary or, ultimately, PSG.

A limitation of this review was limiting to studies that presented data for adults, defined as being 18 years and older. This was decided upon due to the differences in sleep physiology and methods in measuring sleep in children/adolescents; however, this means that the effect of SRC on those younger than 18 years remains unclear. A potential limitation was only examining sports-related concussions. This means that concussions that were sustained during recreational activity or accidents during daily life may not have been included in the literature search. Nevertheless, the relatively robust findings of the included studies, as well as the findings of studies examining moderate to severe TBI, it is likely that non–sports-related concussions would also result in some form of sleep disturbance.

CONCLUSION

The findings of this emergent field of research examining the effect of SRC on sleep indicate that sleep disturbances are prevalent in those who experience an SRC. Because of the lack of consensus on the findings, mainly due to the heterogeneity of the methods of the available studies, it is not possible to provide definitive clinical recommendations specifically related to concussion at this time. Nevertheless, given the role of sleep and the glymphatic system removing amyloid-β and τ-proteins, which are the proteins responsible for CTE, we recommend that sleep is closely monitored after a concussion. If a player who has recently experienced an SRC reports ongoing sleep disturbances and/or excessive daytime sleepiness, it may be necessary to refer the player to a sleep specialist for further clinical investigation and ongoing care. Importantly, more research is needed to properly elucidate the effect of SRC on sleep so that research examining the efficacy of interventions to improve sleep can be undertaken.

All future studies should combine objective and validated subjective sleep assessments, obtain preseason baseline sleep measures, and collect sleep data as soon after the SRC as feasibly possible to ensure that any acute changes in sleep is captured. Furthermore, we propose that screening for underlying mental health and clinical sleep disorders, usage of medication that will affect sleep, and even identifying those with a genetic predisposition to concussion and poor sleep via the APOε4 allele should also occur.

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